ADVANCED DATABASE SYSTEMS

AfolabiPay: Database Project Report

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Project repository: github.com/hafolabi/Oladipo_Afolabi_301919753

1. Introduction

I selected AfolabiPay, a fictional fintech platform, as the case study for this Advanced Database Systems project. It provided a practical way to explore how databases support deposits, withdrawals, transfers, account balances and payment history. The work covers five related activities. PostgreSQL was used for the main financial records, while MongoDB provided a document-based comparison. All customers and payments are fictional. Amounts were recorded in kobo (100 kobo = NGN 1), and balances were calculated from completed transaction entries rather than stored as editable figures.

2. Database Design and Modelling

The relational database was designed around five PostgreSQL tables: customers, accounts, transactions, transaction_entries and audit_events. A customer can have several accounts, and each payment is linked to entries showing the movement of funds. Audit events keep a record of important payment actions.

Primary and foreign keys were used to maintain the relationships between records. The transaction_entries table has a combined key made up of the transaction ID and account ID. Unique constraints, required fields and validation rules help prevent duplicate or incorrect records. The design also follows the principles of normalisation (1NF, 2NF and 3NF) by separating customer, account and payment information and avoiding unnecessary repetition (PostgreSQL Global Development Group, n.d.-a).

Test results: The sample data included 4 customers, 5 accounts, 7 transactions, 10 entries and 8 audit events. The database rejected all nine invalid writes: six tested the original constraints, while three checked customer status and timestamps. The final balances were NGN 8,000 for Ada and NGN 7,000 for Bello. Schema checks ER diagram

3. Query Processing and Optimisation

Four SQL queries were written to retrieve account balances, payment history, daily transfer totals and payment problems. To find out whether history searches could run faster, I tested an index on account ID and transaction ID. The same query and data were used before and after adding the index (PostgreSQL Global Development Group, n.d.-c, n.d.-e).

Measure	Before indexing	After indexing
Payment-history rows	203	203 (same results)
Median query time (five runs)	44.248 ms	1.971 ms
Additional index storage	None	1,256 kB

Adding the index brought the median query time down from 44.248 ms to 1.971 ms, about 22.45 times faster, with the same 203 rows returned. It needed another 1,256 kB of storage and must be maintained as data changes. The query selected only 203 of 40,410 entries; searches returning more rows may benefit less. These results apply to the PGlite 0.5.12 / PostgreSQL 18.6 test environment. Plans and measurements

4. Transactions and Concurrency Control

Transaction tests were carried out on PostgreSQL 18.4 to see what happens when a payment succeeds, fails or overlaps with another request. I used separate database connections to check these situations. Before completing a transfer, the relevant accounts were locked, the available funds were checked, and the payment, its two entries and an audit event were recorded. A successful transfer was committed, while an error caused the transaction to roll back (PostgreSQL Global Development Group, n.d.-d).

Test	Observed result
Commit	A NGN 2,000 transfer reduced Ada's balance from NGN 10,000 to NGN 8,000 and increased Bello's from NGN 0 to NGN 2,000.
Rollback	A deliberate error cancelled the operation. No new transaction, entry or audit event was saved, and balances did not change.
Read Committed	One connection could not see another connection's unfinished payment, but could see it after it was committed.
Concurrent withdrawals	Two requests each attempted to withdraw NGN 800 from NGN 1,000. One succeeded; the other waited, saw NGN 200 remaining and was rejected.

All four checks passed. A separate connection confirmed that the second withdrawal was waiting for a lock. Once it could continue, only NGN 200 remained, so it was rejected. This depends on every payment following the same lock-then-check sequence; direct database writes can bypass it. The shared clearing-account lock also makes deposits and withdrawals wait for each other, even for different customers. Transaction evidence

5. NoSQL and Document-Based Databases

For the NoSQL activity, I created two MongoDB collections: customer_profiles and transaction_events. Customer details were grouped in profile documents, while each payment event stored its two accounting entries and references to the relevant customers. This structure keeps related event information together without requiring table joins (MongoDB, Inc., n.d.-a, n.d.-b).

In the sample transfer TRF-001, account 1000000001 was debited by 200,000 kobo and account 1000000002 was credited by the same amount. BSON Long integers were used to preserve exact money values. The sample database contained two profiles and three payment events. The queries returned customer information by email, an account's history and completed transfers, as intended. MongoDB evidence

MongoDB was useful for organising flexible customer and payment documents, but its references do not provide the same foreign-key checks as PostgreSQL. Saving a single event atomically also does not, on its own, prevent overdrafts across different payments. PostgreSQL therefore remained the main financial record in the proposed system. Automatic synchronisation between the two databases was not implemented.

6. Running the Project

The application can be run from the project repository using npm run setup followed by npm start. Its menu allows sample payments and account-history searches. The README explains the steps, and the saved checks can be repeated using npm run check:p1 through npm run check:p4. Restarting the application resets the sample data.

7. Distributed and Cloud Database Proposal

For the final activity, I proposed a cloud database arrangement with a PostgreSQL primary and standby in separate availability zones, together with three MongoDB replica-set members across three zones. This part remained a design proposal; no cloud deployment or failover test was carried out.

With synchronous PostgreSQL replication, a payment can wait for the standby to save its commit record before being confirmed. This helps protect confirmed payments, but an unavailable standby can hold up new writes even when the primary is still running. During failover, the old primary must be stopped from accepting writes (PostgreSQL Global Development Group, n.d.-b).

MongoDB replication could improve availability when a server fails, but secondary reads may be delayed and writes may pause if a majority of members is unavailable (MongoDB, Inc., n.d.-c, n.d.-d). Payment decisions would therefore continue to use the PostgreSQL primary and the account-locking method tested earlier. Sharding was not proposed for this small dataset because dividing the ledger across servers would add complexity to transfers.

8. Reflection and Conclusion

The withdrawal test shows why the order of the steps matters. Two requests tried to withdraw NGN 800 each from an account holding NGN 1,000. Read Committed alone would not prevent overspending. Locking first meant that the second request checked the balance after the first had finished. The recorded wait and remaining NGN 200 show how this worked.

The design also shows the limits of table constraints. Two entries can each be valid without adding up to a balanced payment. The payment code handles this when saving a transaction, while the checking queries look for problems in stored records. PostgreSQL remains the main financial record, with MongoDB used for the document example. The indexing test makes a similar point: a faster query is useful only if its answers stay correct.

References

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